

Breeti Bandyopadhyay

COMPUTER SCIENCE AND ENGINEERING

226, Dakshin marg, Vidya Vihar, Pilani, Rajasthan • 333031 • bb916@snu.edu.in • 88904 72117
<https://www.linkedin.com/in/breeti-bandyopadhyay-68894b31a/>

EDUCATION

B.Tech Computer Science | 2024-2028

Shiv Nadar University, Noida

- **CGPA: 9.32**
- I am also focused on to complete in minor in Mathematics.
- Dean's List (SNU) (2024-25) - Ranked in the top 10% of the Computer Science cohort

Schooling | 2021-2024

Birla Balika Vidyapeeth, Pilani

- Class X (CBSE): **97.4%** | | Class XII (CBSE): **97%**

EXPERIENCE

Research Intern | Jun 2025 - Jul 2025

5G Use Case Lab, BITS Pilani

- Implemented deep learning models for 5G signal modulation recognition using FFT-based spectral analysis
- Collaborated with EEE faculty on wireless communication research

Research Collaborator | Dec 2024 - Jan 2025

Physics Department, BITS Pilani

- Co-authored DFT study on reversible hydrogen storage in 2D holey graphyne published in Int. J. Hydrogen Energy (**Impact Factor: 8.3**), DOI: <https://doi.org/10.1016/j.ijhydene.2025.150044>
- Conducted VASP simulations for structure optimization and electronic analysis (DOS, band structure); evaluated metrics meeting DOE target

PROJECTS

Face Recognition Photo Organizer: Developed automated photo clustering tool using face_recognition to detect and group individuals across image collections

- Implemented quality-based multi-reference matching with automatic merge detection; integrated CodeFormer for face enhancement
- Stack: Python, face_recognition, OpenCV, NumPy

Movement Video Synchronization Tool: Built pose detection system using MediaPipe to automatically synchronize videos through joint angle comparison and temporal alignment researched and implemented sustainable alternatives to hazardous chemicals

- Implemented angle extraction algorithm (16+ joints) with similarity scoring; integrated FFmpeg pipeline for standardized processing and visualization
- Stack: Python, OpenCV, MediaPipe, NumPy, Ffmpeg

COURSES AND CERTIFICATIONS

- **LinkedIn CoachIn Program (2026):** Selected among top 100 women engineering students nationwide for intensive mentorship in DSA, problem-solving, and tech career prep.
- **Advanced Learning Algorithms + :** by DeepLearning.AI & Stanford University
- **Supervised Machine Learning: Regression and Classification :** by DeepLearning.AI & Stanford Uni.

TECHNICAL SKILLS & INTERESTS

- **Programming:** Python, Java, C
- **ML/CV:** Scikit-Learn, MediaPipe, Face rcognition, Tdnsorflow
- **Tools:** Git, Linux, VS Code, VESTA, VASP

HONOURS AND ACHIEVEMENTS

- **State Topper, Rajasthan:** KAMP NASTA 2023: 1st in state, among 500,000+ participants.
- **School Rank 2:** Class 12th